

Qualification Pack



Aerospace CNC Machinist

QP Code: AAS/Q1001

Version: 3.0

NSQF Level: 4

Aerospace and Aviation Sector Skill Council || No. 11, Dynamatic Park, Peenya
Bangalore 5600058 || email:headops@aassc.in

Qualification Pack

Contents

AAS/Q1001: Aerospace CNC Machinist	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
AAS/N1011: Understanding Accuracy of machining of aerospace components	5
AAS/N1002: Perform pre-machining activities	10
AAS/N1004: Perform post machining activities	16
ASC/N0021: Maintain 5S at the work premises	22
DGT/VSQ/N0102: Employability Skills (60 Hours)	29
Assessment Guidelines and Weightage	36
<i>Assessment Guidelines</i>	36
<i>Assessment Weightage</i>	37
Acronyms	38
Glossary	39

Qualification Pack

AAS/Q1001: Aerospace CNC Machinist

Brief Job Description

This role primarily involves CNC machining of aerospace components/ structures in all pre-machining & post machining stages alongwith self inspection & gauging.

Personal Attributes

The individual should have basic communication ability and should be responsible in order to maintain procedures. He should have the ability to understand engineering drawings, dos and donts of manufacturing process defined in work instruction or by a shop supervisor.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [AAS/N1011: Understanding Accuracy of machining of aerospace components](#)
2. [AAS/N1002: Perform pre-machining activities](#)
3. [AAS/N1003: Perform machining operations](#)
4. [AAS/N1004: Perform post machining activities](#)
5. [ASC/N0021: Maintain 5S at the work premises](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Aerospace and Aviation
Sub-Sector	Manufacturing & Assembly
Occupation	Machining
Country	India
NSQF Level	4
Credits	21
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7223.5001

Qualification Pack

Minimum Educational Qualification & Experience	12th grade Pass (or Equivalent) OR 10th grade pass with 3 Years of experience OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	3.0
Reference code on NQR	QG-04-AA-04190-2025-V2-AASSC
NQR Version	3

Qualification Pack

AAS/N1011: Understanding Accuracy of machining of aerospace components

Description

This module provides Aerospace CNC Machinists with the foundational and applied knowledge required to achieve high accuracy in machining aerospace components. It covers dimensional tolerances, surface finish requirements, use of precision measuring instruments, and methods for maintaining quality in CNC machining processes.

Scope

The scope covers the following :

- Ensure high-accuracy machining of aerospace components by interpreting drawings, tolerances, and surface finish requirements.
- Operate CNC machines with precision, monitor machining parameters, and apply corrective actions to maintain dimensional accuracy.
- Perform accurate measurement and inspection using calibrated instruments to verify compliance with aerospace quality standards.

Elements and Performance Criteria

Interpret Machining Accuracy Requirements

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the importance of machining accuracy in aerospace components and its effect on airworthiness and system performance.
- PC2.** Interpret engineering drawings, GD&T (Geometric Dimensioning & Tolerancing), and process plans to determine accuracy requirements.
- PC3.** Identify dimensional tolerances and surface finish specifications for aerospace parts.
- PC4.** Understand the impact of incorrect settings, tool wear, and fixture alignment on final dimensions.

Apply Accuracy Control Techniques in CNC Machining

To be competent, the user/individual on the job must be able to:

- PC5.** Set up and operate CNC machines as per specifications ensuring control over machining accuracy.
- PC6.** Monitor machining parameters (feed, speed, depth of cut) that influence part tolerance.
- PC7.** Check for thermal distortion, vibration, and tool deflection during machining.
- PC8.** Take corrective actions in case of dimensional deviations or tool-related issues.

Performing Accurate Measurement and Inspection

To be competent, the user/individual on the job must be able to:

- PC9.** Select appropriate measurement instruments such as micrometers, height gauges, bore gauges, and CMMs.
- PC10.** Calibrate measurement instruments before use.
- PC11.** Measure finished parts against tolerance limits and record results accurately.

Qualification Pack

PC12. Report any non-conformities to supervisors or the quality control team.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The significance of machining accuracy in aerospace manufacturing and maintenance.
- KU2.** Aerospace tolerancing standards and how to read GD&T symbols.
- KU3.** Causes of inaccuracies in CNC machining (e.g., backlash, thermal expansion, worn tools).
- KU4.** Functioning and use of precision measuring tools and gauges.
- KU5.** Calibration methods and importance of traceability in measurement.
- KU6.** Procedures for documenting measurements and identifying non-conformances.
- KU7.** Importance of quality control and continuous improvement in aerospace machining.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret technical drawings and inspection standards.
- GS2.** Maintain accurate records of part measurements and deviations.
- GS3.** Communicate clearly with the quality assurance and production teams.
- GS4.** Follow SOPs and safety guidelines related to precision machining and measurement.
- GS5.** Identify and correct basic machine setup errors affecting part accuracy.
- GS6.** Analyze tolerance stack-up and suggest process improvements.
- GS7.** Maintain a clean, controlled work environment to avoid contamination of parts and tools.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Interpret Machining Accuracy Requirements</i>	15	20	-	-
PC1. Understand the importance of machining accuracy in aerospace components and its effect on airworthiness and system performance.	4	5	-	-
PC2. Interpret engineering drawings, GD&T (Geometric Dimensioning & Tolerancing), and process plans to determine accuracy requirements.	4	5	-	-
PC3. Identify dimensional tolerances and surface finish specifications for aerospace parts.	4	5	-	-
PC4. Understand the impact of incorrect settings, tool wear, and fixture alignment on final dimensions.	3	5	-	-
<i>Apply Accuracy Control Techniques in CNC Machining</i>	15	20	-	-
PC5. Set up and operate CNC machines as per specifications ensuring control over machining accuracy.	4	5	-	-
PC6. Monitor machining parameters (feed, speed, depth of cut) that influence part tolerance.	4	5	-	-
PC7. Check for thermal distortion, vibration, and tool deflection during machining.	4	5	-	-
PC8. Take corrective actions in case of dimensional deviations or tool-related issues.	3	5	-	-
<i>Performing Accurate Measurement and Inspection</i>	10	20	-	-
PC9. Select appropriate measurement instruments such as micrometers, height gauges, bore gauges, and CMMs.	3	5	-	-
PC10. Calibrate measurement instruments before use.	3	5	-	-
PC11. Measure finished parts against tolerance limits and record results accurately.	2	5	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Report any non-conformities to supervisors or the quality control team.	2	5	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1011
NOS Name	Understanding Accuracy of machining of aerospace components
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Machining
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

AAS/N1002: Perform pre-machining activities

Description

This NOS unit describes the knowledge, understanding and skills required of an individual to carry out pre-machining activities of aerospace components on CNC machining centre

Scope

The scope covers the following :

- Understand and interpret manufacturing drawings, process sheets, and CNC setup instructions for aerospace components.
- Prepare tools, jigs, fixtures, and machine references as per the specified machining requirements.
- Ensure machine readiness, perform trial runs if required, and coordinate with engineers or supervisors to confirm technical accuracy before machining.

Elements and Performance Criteria

Collect manufacturing drawing and process requirements

To be competent, the user/individual on the job must be able to:

- PC1.** obtain drawings, route card / work instruction / CNC process sheet / CNC set-up sheet, details of fixtures (if any) from shop supervisor
- PC2.** check right Computer Numerically Controlled (CNC) programme and right tools required for machining of aerospace components
- PC3.** draw the raw-material as per specification
- PC4.** check for satisfactory functioning and calibration of the machine
- PC5.** plan the machining process as per process sheet or route card

Prepare tools, arrange jigs and fixture

To be competent, the user/individual on the job must be able to:

- PC6.** arrange the required tool from stores
- PC7.** arrange appropriate jigs and fixtures as mentioned in the manufacturing drawings / route card / setup sheet / process sheet
- PC8.** establish machine datum reference after following proper procedures as per CNC set up sheet
- PC9.** check output requirements/ limits of machining e.g. surface finish, specific orientation, gauge inspection etc.
- PC10.** understand any other specific requirement for machining from process sheet/ route card or work instruction and grain direction, if any

Pre-machining readiness

To be competent, the user/individual on the job must be able to:

- PC11.** ensure that dos and don'ts provided in the work instruction is adhered
- PC12.** trial run the CNC programme if specified by the CNC programmer or provided route card / work instruction
- PC13.** discuss technical matters related to machine programming with the engineer/ supervisor

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** different types of products manufactured by the company
- KU3.** different types of machining processes/ tools available
- KU4.** aerospace manufacturing standard and procedures such as ISO 9001, AS9100, etc
- KU5.** different types of machining processes
- KU6.** basic fundamentals of machines and mechanics
- KU7.** different types of tools used in the machining process with respect to type of process to be conducted
- KU8.** basic principles of 5S in manufacturing
- KU9.** application of coolant and lubricants
- KU10.** impact of various machining processes on the final product outcome
- KU11.** basic arithmetic and calculation methods for tolerance limits
- KU12.** safety guidelines related to different machines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write basic level notes and observations
- GS2.** draw basic level geometrical/ mechanical drawings and charts
- GS3.** read and interpret workplace related documentation
- GS4.** read and interpret engineering drawings and sketches to understand the dimension of the output product
- GS5.** discuss task lists and job requirements with co-workers
- GS6.** effectively communicate information to team members
- GS7.** discuss with supervisor in order to understand the nature of the problem
- GS8.** attentively listen with full attention and comprehend the information given by the speaker
- GS9.** judge when to ask for help from a supervisor
- GS10.** suggest options to operators in case any issue is observed during operations
- GS11.** plan work assigned on a daily basis and provide estimates of time required for each piece of work
- GS12.** prioritise actions to achieve required outcomes
- GS13.** ensure that customer needs are assessed and every effort is made to provide satisfactory service.
- GS14.** recognise a work place problem or a potential problem and take action
- GS15.** determine problems needing priority action
- GS16.** gather information and provide assistance as required to solve problem
- GS17.** refer problems outside area of responsibility to approach

Qualification Pack

- GS18.** analyse the complexity of work to determine if it can be successfully carried out
- GS19.** learn from mistakes by analysing and discussing with peers/ seniors
- GS20.** discuss new ideas and participate in new initiatives
- GS21.** follow instructions and work on areas of improvement identified
- GS22.** complete the assigned tasks with minimum supervision
- GS23.** complete the job defined by the supervisor within the timelines and quality norms
- GS24.** concentrate on task at hand and complete it without errors
- GS25.** apply balanced judgments to different situations
- GS26.** concentrate on task at hand and complete it without errors
- GS27.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Collect manufacturing drawing and process requirements</i>	16	20	-	-
PC1. obtain drawings, route card / work instruction / CNC process sheet / CNC set-up sheet, details of fixtures (if any) from shop supervisor	3	4	-	-
PC2. check right Computer Numerically Controlled (CNC) programme and right tools required for machining of aerospace components	3	4	-	-
PC3. draw the raw-material as per specification	3	4	-	-
PC4. check for satisfactory functioning and calibration of the machine	3	4	-	-
PC5. plan the machining process as per process sheet or route card	4	4	-	-
<i>Prepare tools, arrange jigs and fixture</i>	20	20	-	-
PC6. arrange the required tool from stores	4	4	-	-
PC7. arrange appropriate jigs and fixtures as mentioned in the manufacturing drawings / route card / setup sheet / process sheet	4	4	-	-
PC8. establish machine datum reference after following proper procedures as per CNC set up sheet	4	4	-	-
PC9. check output requirements/ limits of machining e.g. surface finish, specific orientation, gauge inspection etc.	4	4	-	-
PC10. understand any other specific requirement for machining from process sheet/ route card or work instruction and grain direction,if any	4	4	-	-
<i>Pre-machining readiness</i>	12	12	-	-
PC11. ensure that dos and donts provided in the work instruction is adhered	4	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. trial run the CNC programme if specified by the CNC programmer or provided route card / work instruction	4	4	-	-
PC13. discuss technical matters related to machine programming with the engineer/ supervisor	4	4	-	-
NOS Total	48	52	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1002
NOS Name	Perform pre-machining activities
Sector	Aerospace and Aviation
Sub-Sector	Manufacturing & Assembly
Occupation	Machining
NSQF Level	4
Credits	5
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

AAS/N1004: Perform post machining activities

Description

This NOS unit is about conducting all post machining operations such as performing minor maintenance, assisting in tool change operations and de-burring activities

Scope

The scope covers the following :

- Conduct daily and minor machine maintenance (cleaning, oiling, coolant refilling, etc.)
- Perform de-burring using hand tools, grinders, or shot blasting methods
- Check the quality of machined components using measuring instruments
- Assist in and carry out tool change operations safely and correctly
- Maintain cleanliness, safety, and compliance with operational procedures

Elements and Performance Criteria

Perform minor machine maintenance activities

To be competent, the user/individual on the job must be able to:

- PC1.** maintain the machine as per proper operational condition/ perform daily maintenance check
- PC2.** perform minor machine maintenance activities such as oiling or cleaning machine and its components as per the schedules given in the maintenance plan
- PC3.** clean the hydraulic tank/ gauge/ tools/ fixtures as per the cleaning schedule and the process mentioned in the work instruction/ SOP manual
- PC4.** add coolant and lubricant in the machine reservoir as per the SOPs
- PC5.** remove chips from different machine areas and dispose of scrap or waste material into the disposal area in accordance with the company policies and environmental regulations
- PC6.** perform minor repairs and adjustments to the machine and notify the supervisor/ maintenance team when major service/ repair is required
- PC7.** return drawings, tools and other measuring equipment to respective department

Perform de-burring activity on the machined components

To be competent, the user/individual on the job must be able to:

- PC8.** with the help of the correct tool remove extra burrs, sharp edges, rust and chips from the metal surface
- PC9.** use files, hand grinders, wire brushes, or power tools for performing de-burring operations and ensure usage of Personal Protective Equipment (PPEs) such as eye glasses and hand gloves
- PC10.** trim, scrape, or deburr objects or parts using chisels, scrapers, and other hand tools and equipment
- PC11.** perform shot blasting/ vibro processes for completing de-burring operations (if required)

Check quality of machined component

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC12.** measure the specifications of the finished component and verify conformance as per control plan/ work instruction
- PC13.** use devices like micrometers, vernier calipers, gauges, rulers and any other inspection equipment for measuring specifications with valid calibration status
- PC14.** note down the observations of the basic inspection process and identify pieces which comply with the specified standards
- PC15.** separate the defective pieces into two categories pieces which can be repaired/ modified and pieces which are beyond repair, and maintain records of each category

Tool Changing Process

To be competent, the user/individual on the job must be able to:

- PC16.** ensure that any blunt tool is timely and safely replaced by a new tool
- PC17.** replace machine part as per work instructions, using hand tools or notify supervisor/ engineering personnel for taking corrective actions
- PC18.** observe the tool change cycle in order to ensure that the selected tool is transferred to the spindle from magazine after the previous tool is transferred to the magazine from the spindle
- PC19.** ensure that the zero offset value is chosen at the time of tool changing process

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** different types of products manufactured by the company
- KU3.** different types of machining processes
- KU4.** different types of tools used in the machining process and de-burring process
- KU5.** basic principles of 5S in manufacturing
- KU6.** aerospace manufacturing standard and procedures such as ISO 9001, AS9100, etc
- KU7.** post machining processes like deburring
- KU8.** impact of presence of burrs, edges, chips on the final product performance
- KU9.** the application of coolant and lubricants
- KU10.** basic arithmetic and calculation methods

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write basic level notes and observations
- GS2.** draw basic level drawings and charts
- GS3.** read documents and notes
- GS4.** interpret/ comprehend the information given in the documents and notes
- GS5.** read and interpret symbols given on equipment and work area
- GS6.** discuss task lists and job requirements with co-workers
- GS7.** effectively communicate information to team members

Qualification Pack

- GS8.** question operator/ supervisor in order to understand the nature of the problem
- GS9.** attentively listen with full attention and comprehend the information given by the speaker
- GS10.** judge when to ask for help from a supervisor
- GS11.** suggest options to operators in case any issue is observed during operations
- GS12.** plan work assigned on a daily basis and provide estimates of time required for each piece of work
- GS13.** prioritise actions to achieve required outcomes
- GS14.** ensure that customer needs are assessed and every effort is made to provide satisfactory service
- GS15.** recognise a work place problem or a potential problem and take action
- GS16.** determine problems needing priority action
- GS17.** gather information and provide assistance as required to solve problem
- GS18.** refer problems outside area of responsibility to approach person
- GS19.** analyse the complexity of work to determine if it can be successfully carried out
- GS20.** ability to visualise the final product from the engineering drawing/ machine drawing, sketch provided by the supervisor
- GS21.** analyse the cause of defects related to e.g. cutting tools, machine, fixtures etc
- GS22.** learn from mistakes by analysing and discussing with peers/ seniors
- GS23.** discuss new ideas and participate in new initiatives
- GS24.** follow instructions and work on areas of improvement identified
- GS25.** complete the assigned tasks with minimum supervision
- GS26.** complete the job defined by the supervisor within the timelines and

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform minor machine maintenance activities</i>	17	28	-	-
PC1. maintain the machine as per proper operational condition/ perform daily maintenance check	2	4	-	-
PC2. perform minor machine maintenance activities such as oiling or cleaning machine and its components as per the schedules given in the maintenance plan	2	4	-	-
PC3. clean the hydraulic tank/ gauge/ tools/ fixtures as per the cleaning schedule and the process mentioned in the work instruction/ SOP manual	2	4	-	-
PC4. add coolant and lubricant in the machine reservoir as per the SOPs	2	4	-	-
PC5. remove chips from different machine areas and dispose of scrap or waste material into the disposal area in accordance with the company policies and environmental regulations	3	4	-	-
PC6. perform minor repairs and adjustments to the machine and notify the supervisor/ maintenance team when major service/ repair is required	3	4	-	-
PC7. return drawings, tools and other measuring equipment to respective department	3	4	-	-
<i>Perform de- burring activity on the machined components</i>	12	16	-	-
PC8. with the help of the correct tool remove extra burrs, sharp edges, rust and chips from the metal surface	3	4	-	-
PC9. use files, hand grinders, wire brushes, or power tools for performing de-burring operations and ensure usage of Personal Protective Equipment (PPEs) such as eye glasses and hand gloves	3	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. trim, scrape, or deburr objects or parts using chisels, scrapers, and other hand tools and equipment	3	4	-	-
PC11. perform shot blasting/ vibro processes for completing de-burring operations (if required)	3	4	-	-
<i>Check quality of machined component</i>	12	16	-	-
PC12. measure the specifications of the finished component and verify conformance as per control plan/ work instruction	3	4	-	-
PC13. use devices like micrometers, vernier calipers, gauges, rulers and any other inspection equipment for measuring specifications with valid calibration status	3	4	-	-
PC14. note down the observations of the basic inspection process and identify pieces which comply with the specified standards	3	4	-	-
PC15. separate the defective pieces into two categories pieces which can be repaired/ modified and pieces which are beyond repair, and maintain records of each category	3	4	-	-
<i>Tool Changing Process</i>	12	16	-	-
PC16. ensure that any blunt tool is timely and safely replaced by a new tool	3	4	-	-
PC17. replace machine part as per work instructions, using hand tools or notify supervisor/ engineering personnel for taking corrective actions	3	4	-	-
PC18. observe the tool change cycle in order to ensure that the selected tool is transferred to the spindle from magazine after the previous tool is transferred to the magazine from the spindle	3	4	-	-
PC19. ensure that the zero offset value is chosen at the time of tool changing process	3	4	-	-
NOS Total	53	76	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1004
NOS Name	Perform post machining activities
Sector	Aerospace and Aviation
Sub-Sector	Manufacturing & Assembly
Occupation	Machining
NSQF Level	4
Credits	5
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

ASC/N0021: Maintain 5S at the work premises

Description

This NOS is about ensuring all 5 S activities both at the shop floor and the office area to facilitate increase in work productivity

Elements and Performance Criteria

Ensure sorting

To be competent, the user/individual on the job must be able to:

- PC1..** follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.
- PC2..** ensure segregation of waste in hazardous/ non hazardous waste as per the sorting work instructions
- PC3..** follow the technique of waste disposal and waste storage in the proper bins as per sop
- PC4..** segregate the items which are labelled as red tag items for the process area and keep them in the correct places
- PC5.** sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5s guidelines/ work instructions
- PC6. .** ensure that areas of material storage areas are not overflowing
- PC7.** properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required
- PC8.** return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area
- PC9.** follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards
- PC10.** follow the proper labeling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists

Ensure proper documentation and storage (organizing , streamlining)

To be competent, the user/individual on the job must be able to:

- PC11.** check that the items in the respective areas have been identified as broken or damaged
- PC12.** follow the given instructions and check for labelling of fluids, oils. lubricants, solvents, chemicals etc. and proper storage of the same to avoid spillage, leakage, fire etc
- PC13.** make sure that all material and tools are stored in the designated places and in the manner indicated in the 5s instructions

Ensure cleaning of self and the work place

To be competent, the user/individual on the job must be able to:

- PC14.** check whether safety glasses are clean and in good condition
- PC15.** keep all outside surfaces of recycling containers are clean
- PC16..** ensure that the area has floors swept, machinery clean and generally clean. in case of cleaning, ensure that proper displays are maintained on the floor which indicate potential safety hazards

Qualification Pack

- PC17..** check whether all hoses, cabling & wires are clean, in good condition and clamped to avoid any mishap or mix up
- PC18..** ensure workbenches and work surfaces are clean and in good condition
- PC19.** follow the cleaning schedule for the lighting system to ensure proper illumination
- PC20.** store the cleaning material and equipment in the correct location and in good condition
- PC21.** ensure self-cleanliness - clean uniform, clean shoes, clean gloves, clean helmets, personal hygiene

Ensure sustenance

To be competent, the user/individual on the job must be able to:

- PC22.** follow the daily cleaning standards and schedules to create a clean working environment
- PC23.** attend all training programs for employees on 5 s
- PC24.** support the team during the audit of 5 s
- PC25.** participate actively in employee work groups on 5s and encourage team members for active participation
- PC26.** follow the guidelines for what to do and what not to do to build sustainability in 5s as mentioned in the 5s check lists/ work instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards, procedures and policies related to 5S followed in the company
- KU2.** have basic knowledge of 5S procedures
- KU3.** know various types 5s practices followed in various areas
- KU4.** understand the 5S checklists provided in the department/ team
- KU5.** have skills to identify useful & non useful items
- KU6.** have knowledge of labels , signs & colours used as indicators
- KU7.** knowledge on how to sort and store various types of tools, equipment, material etc.
- KU8.** know , how to identify various types of waste products
- KU9.** understand the impact of waste/ dirt/ dust/unwanted substances on the process/ environment/ machinery/ human body
- KU10.** have knowledge of best ways of cleaning & waste disposal
- KU11.** understand the importance of standardization in processes
- KU12.** understand the importance of sustainability in 5S
- KU13.** have knowledge of TQM process
- KU14.** have knowledge of various materials and storage norms
- KU15.** understand visual controls, symbols, graphs etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write basic level notes and observations

Qualification Pack

- GS2.** note down observations (if any) related to the process
- GS3.** read 5S instructions put up across the plant premises
- GS4.** effectively communicate information to team members inform employees in the plant and concerned functions about 5S
- GS5.** question the process head in order to understand the 5S related issues
- GS6.** attentively listen with full attention and comprehend the information given by the speaker during 5S training programs
- GS7.** use common sense and make judgments during day to day basis
- GS8.** use reasoning skills to identify and resolve basic problems using 5S
- GS9.** persuade co team members to follow 5 S
- GS10.** ensure that the co team members understand the importance of using 5 S tool
- GS11.** use innovative skills to perform and manage 5 S activities at the work desk and the shop floor
- GS12.** exhibit inquisitive behaviour to seek feedback and question on the existing set patterns of work
- GS13.** do what is right, not what is a popular practices
- GS14.** follow shop floor rules& regulations and avoid deviations; make 5S an integral way of life
- GS15.** ensure self-cleanliness on a daily basis
- GS16.** demonstrate the will to keep the work area in a clean and orderly manner

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure sorting</i>	10	30	-	-
PC1.. follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.	1	3	-	-
PC2.. ensure segregation of waste in hazardous/ non hazardous waste as per the sorting work instructions	1	3	-	-
PC3.. follow the technique of waste disposal and waste storage in the proper bins as per sop	1	3	-	-
PC4.. segregate the items which are labelled as red tag items for the process area and keep them in the correct places	1	3	-	-
PC5. sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5s guidelines/ work instructions	1	3	-	-
PC6. . ensure that areas of material storage areas are not overflowing	1	3	-	-
PC7. properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required	1	3	-	-
PC8. return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area	1	3	-	-
PC9. follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards	1	3	-	-
PC10. follow the proper labeling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists	1	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure proper documentation and storage (organizing , streamlining)</i>	3	9	-	-
PC11. check that the items in the respective areas have been identified as broken or damaged	1	3	-	-
PC12. follow the given instructions and check for labelling of fluids, oils. lubricants, solvents, chemicals etc. and proper storage of the same to avoid spillage, leakage, fire etc	1	3	-	-
PC13. make sure that all material and tools are stored in the designated places and in the manner indicated in the 5s instructions	1	3	-	-
<i>Ensure cleaning of self and the work place</i>	8	24	-	-
PC14. check whether safety glasses are clean and in good condition	1	3	-	-
PC15. keep all outside surfaces of recycling containers are clean	1	3	-	-
PC16.. ensure that the area has floors swept, machinery clean and generally clean. in case of cleaning, ensure that proper displays are maintained on the floor which indicate potential safety hazards	1	3	-	-
PC17.. check whether all hoses, cabling & wires are clean, in goodcondition and clamped to avoid any mishap or mix up	1	3	-	-
PC18.. ensure workbenches and work surfaces are clean and in good condition	1	3	-	-
PC19. follow the cleaning schedule for the lighting system to ensure proper illumination	1	3	-	-
PC20. store the cleaning material and equipment in the correct location and in good condition	1	3	-	-
PC21. ensure self-cleanliness - clean uniform, clean shoes, clean gloves, clean helmets, personal hygiene	1	3	-	-
<i>Ensure sustenance</i>	4	12	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. follow the daily cleaning standards and schedules to create a clean working environment	1	3	-	-
PC23. attend all training programs for employees on 5 s	0.5	2	-	-
PC24. support the team during the audit of 5 s	1	3	-	-
PC25. participate actively in employee work groups on 5s and encourage team members for active participation	0.5	2	-	-
PC26. follow the guidelines for what to do and what not to do to build sustainability in 5s as mentioned in the 5s check lists/ work instructions	1	2	-	-
NOS Total	25	75	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N0021
NOS Name	Maintain 5S at the work premises
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
AAS/N1011.Understanding Accuracy of machining of aerospace components	40	60	-	-	100	14
AAS/N1002.Perform pre-machining activities	48	52	-	-	100	22
AAS/N1003.Perform machining operations	40	60	-	-	100	22
AAS/N1004.Perform post machining activities	53	76	-	-	129	22
ASC/N0021.Maintain 5S at the work premises	25	75	-	-	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	226	353	-	-	579	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
CNC	Computer Numerical Control
GD&T	Geometric Dimensioning and Tolerancing
CMMs	Coordinate Measuring Machines

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.

Qualification Pack

National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack(QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique Qualification pack code.
Unit Code	Unit code is a unique identifier for an occupational standard, which is denoted by 'N'.
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.